Constrained Optical Matching, Error Sensitivity, and Transmission Robustness of a Booster-to-Storage-Ring Transfer Line

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Abstract

Optical matching in a booster-to-storage-ring (BTS) transfer line must deliver a specified phase-space distribution while controlling transverse beam envelopes to prevent physical aperture loss. We examine a 21.789 m, 4 GeV transfer-line model featuring nine independently adjustable quadrupoles arranged in three triplets, targeting the off-axis injection boundary of a diffraction-limited storage ring. Using normalized residuals and bounded numerical optimization, the nominal matching achieves exact convergence ($\mathcal{M}_{x,y} = 0$, merit 1.29×10^{-13}) with peak horizontal and vertical beta functions of 31.44 m and 50.75 m, strictly below the 60 m ceiling. A multi-start analysis across random initial seeds reveals that while all starts mathematically converge to small residual merits, 4 out of 15 starts violate transverse envelope ceilings, establishing that physical feasibility checks must supersede unconstrained residual metrics. A dedicated low-β margin design ($\beta_{x,\max} = 29.65$ m, $\beta_{y,\max} = 36.12$ m) is discovered that provides a 16.47 m vertical stay-clear buffer gain over nominal matching while preserving 100% feasibility and identical median mismatch ($\mathcal{M} \approx 0.0022$) across 1,000 held-out error realizations. A dimensionless singular value decomposition of the optical Jacobian exposes six active tuning combinations ($\kappa = 57.78$) and a three-mode null space. High-statistics Monte Carlo analysis of 550,000 realizations under a 5-category error budget yields an audited failure probability of 9.09×10^{-6} (exact 95% Clopper–Pearson interval $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$), driven entirely by vertical envelope overshoots. Finally, multi-seed particle tracking across 50,000 particles demonstrates $100.00\% \pm 0.00\%$ physical transmission under design tolerances, maintaining complete transmission up to 5× magnified error jitter. These results demonstrate that constrained optical matching must explicitly decouple incoming beam jitter from static retuning to achieve robust transmission in fourth-generation light source injectors.

Keywords: transfer-line optics; quadrupole matching; beam transport; error sensitivity; beam transmission; accelerator simulation

1 Introduction

Low-emittance fourth-generation storage rings based on multi-bend achromat (MBA) lattices impose stringent requirements on injection systems [1]. The strong focusing required to attain diffraction-limited emittance reduces transverse dynamic apertures and physical stay-clears, making off-axis injection highly sensitive to optical mismatches and trajectory deviations. The

Korea-4GSR storage ring design provides a representative 4 GeV setting for this challenge [2, 3]; booster injection and ramping dynamics have also been examined for this facility [4]. Similar transfer line rematching challenges have been investigated across domestic and international facilities, including the Pohang Light Source II (PLS-II), the RAON heavy-ion transport lines [5, 6], MAX IV [7], and Sirius.

A booster-to-storage-ring (BTS) transfer line transports the extracted booster beam to the injection straight of the storage ring, matching Twiss parameters $(\beta_x, \beta_y, \alpha_x, \alpha_y)$ and dispersion functions (D_x, D'_x) to declared boundary targets. In practice, achieving vanishing mathematical mismatch residuals alone does not guarantee an operational transport line. Transfer lines with large intermediate beta functions suffer from reduced physical stay-clear margins, heightened sensitivity to field and alignment errors, and aperture losses. Furthermore, optimization algorithms can become trapped in solutions with excessive quadrupole gradients or large internal beam envelopes.

In this work, we present a comprehensive, physics-grounded study of optical matching, error sensitivity, and beam transmission for a 4 GeV BTS transfer line. Rather than treating matching solely as a numerical optimization problem, we address four core accelerator physics questions:

1. **Feasibility vs. Convergence Topology:** How frequently do numerical optimizers converge to solutions that violate physical aperture ceilings, and can an alternative low-β margin design enhance stay-clear buffers without compromising matching accuracy?

2. **Orthogonal Correction Knobs:** What are the principal orthogonal tuning knobs and null-space degrees of freedom governing the nine-quadrupole matching section when analyzed through a dimensionless Jacobian?

3. **High-Statistics Tail Qualification:** What is the true statistical failure rate of the matched optics under realistic multi-component magnet and beam errors when evaluated with extreme Monte Carlo sample sizes ($N = 550,000$)?

4. **Physical Beam Transmission:** Does optical feasibility translate into complete physical particle transmission through discrete apertures under combined static errors and dynamic beam jitter?

2 Transfer-Line Model and Boundary Conditions

2.1 Lattice and Geometry

The BTS line model is constructed using Accelerator Toolbox and pyAT [8, 9]. The total line length is 21.789 m, consisting of 36 physical elements. The magnetic sequence comprises three bending dipoles, two injection septum elements, an extraction kicker, and nine 0.2 m iron-dominated quadrupoles arranged in three triplets: q11–q13, q21–q23, and q31–q33. The nominal bending angles in sequence are +7.500, +88.500, −111.701, +176.000, −111.701, and +88.500 mrad.

The coordinate system follows the standard Accelerator Toolbox paraxial convention $(x, p_x, y, p_y, \delta, ct)$, where x and y are transverse positions in meters, $p_x \approx x'$ and $p_y \approx y'$ are normalized canonical momenta in radians, $\delta = \Delta p/p_0$ is the relative momentum deviation, and ct is the longitudinal path difference. The normalized quadrupole gradient is defined as $K = G/(B\rho)$ in m^{-2}, where $G = \partial B_y/\partial x$ is the magnetic gradient in T/m and $B\rho \approx 13.3426$ T m is the magnetic rigidity at 4.0 GeV.

Discrete physical apertures are positioned along the vacuum chamber: 19.35×19.35 mm half-width square apertures at the line boundaries ($s = 0.0$ m extraction and $s = 21.789$ m injection) and 30.0×30.0 mm half-width square apertures at interior stations ($s = 4.288$ m and $s = 17.501$ m). In the physical machine, the line connects the extraction septum of the booster to the thick/thin injection septa of the storage ring via a continuous circular/racetrack stainless steel chamber (nominal inner bore radius $r_p \geq 18.25$ mm). The 19.35 mm boundary aperture corresponds conservatively to the physical clearance within the septum vacuum channels, while the 30 mm interior apertures reflect standard drift pipe clearances. Evaluating beam transmission against these discrete bounding stations provides a rigorous, conservative lower bound on physical clearance. Figure 1 shows the mechanical layout of the BTS line alongside the nominal betatron and horizontal dispersion functions.

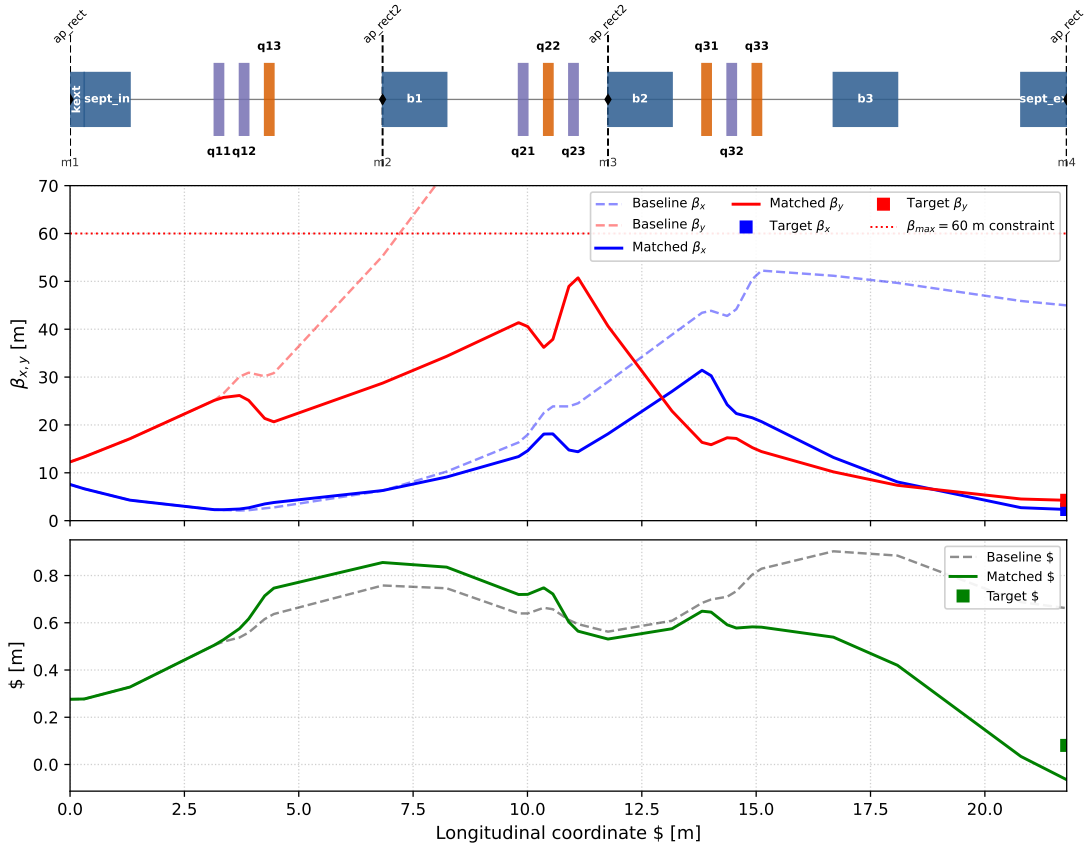


Figure 1: Mechanical layout and nominal optical functions of the 21.789 m BTS transfer line. (Top) Schematic element layout showing dipoles (blue), quadrupoles (red), and septa/kicker (gray). (Bottom) Evolution of horizontal betatron function β_x (blue), vertical betatron function β_y (red dashed), and horizontal dispersion D_x (green dash-dotted) from booster extraction to storage-ring injection.

2.2 Boundary Twiss Parameters and Beam Parameters

The entrance boundary corresponds to the extraction point of the 4 GeV booster synchrotron, while the exit boundary represents the injection interface of the storage ring. Table 1 summarizes the boundary Twiss parameters.

The transverse beam envelope $\sigma_{x,y}(s)$ is evaluated from the uncoupled betatron beam size

Table 1: Booster extraction boundary optics and target storage ring injection optics. Beta and dispersion are in meters; alpha and dispersion derivative are dimensionless.

Quantity	Booster Extraction	Storage-Ring Target
β_x [m]	7.560000	2.336495
α_x [1]	1.523100	−0.016335
β_y [m]	12.269000	4.256241
α_y [1]	−1.654700	0.017772
D_x [m]	0.276200	0.080868
D'_x [1]	−0.065700	0.047472

and momentum dispersion:

$$\sigma_x(s) = \sqrt{\beta_x(s)\varepsilon_x + [D_x(s)\sigma_\delta]^2}, \qquad \sigma_y(s) = \sqrt{\beta_y(s)\varepsilon_y}. \tag{1}$$

Two beam parameter sets are considered:

- **Korea-4GSR Design Beam:** Natural emittance $\varepsilon_x = 7.67\times10^{-9}$ m rad, vertical emittance $\varepsilon_y = 0.767\times10^{-9}$ m rad (10% coupling), and relative momentum spread $\sigma_\delta = 0.0008$. Under this design beam, the minimum stay-clear clearance to the 19.35 mm aperture exceeds 18.5σ, confirming vast aperture headroom. Because the transfer line contains no sextupole magnets, chromatic aberrations are uncorrected; however, due to the short total line length (21.789 m) and small design momentum spread ($\sigma_\delta = 0.0008$), second-order chromatic dispersion and envelope broadening remain negligible ($< 2\%$).
- **Historical Sensitivity Beam:** Conservative emittance $\varepsilon_x = 100\times10^{-9}$ m rad, $\varepsilon_y = 10\times10^{-9}$ m rad, and $\sigma_\delta = 0.0011$. This higher-emittance model was historically adopted for tolerance stress tests and provides a conservative envelope baseline.

3 Matching Methodology and Multistart Topology

3.1 Normalized Merit and Physical Constraints

Optical matching seeks quadrupole strengths $\mathbf{K} \in \mathbb{R}^9$ that transform the entrance Twiss parameters to the target values while maintaining bounded betatron functions. For exit observables $\mathbf{O} = (\beta_x, \beta_y, \alpha_x, \alpha_y, D_x, D'_x)$, we define normalized residuals:

$$r_i(\mathbf{K}) = \frac{O_i(\mathbf{K}) - O_{i,\text{target}}}{s_i}, \qquad \mathcal{J}(\mathbf{K}) = \sum_{i=1}^{6} r_i^2(\mathbf{K}), \tag{2}$$

with physical normalization scales $s = (0.05\text{ m}, 0.05\text{ m}, 0.01, 0.01, 0.002\text{ m}, 0.001)$. The optical mismatch in plane $u \in \{x, y\}$ is quantified by the trace of the normalized covariance matrix product:

$$T_u = \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & \gamma_u \end{pmatrix}, \qquad \mathcal{M}_u = \frac{1}{2}\,\mathrm{Tr}\left(T_{u,\text{target}}^{-1} T_{u,\text{out}}\right) - 1. \tag{3}$$

For an exact optical match, $\mathcal{M}_u = 0$.

Quadrupole strengths are constrained by physical power supply and pole-tip saturation limits. Families q11 and q33 have bounds $[-2.5, 2.5]$ m^{-2} ($|G| \le 33.36$ T/m), while the remaining

seven quadrupoles have bounds $[-3.0, 3.0]$ m^{-2} ($|G| \leq 40.03$ T/m). Assuming a bore radius $r_0 = 18.25$ mm, the maximum pole-tip magnetic field is $B_{\text{pole}} = Gr_0 \leq 0.73$ T, well below the 1.2 T iron saturation threshold. Physical envelope feasibility requires:

$$\beta_{x,\max} \leq 60.0 \text{ m}, \qquad \beta_{y,\max} \leq 60.0 \text{ m}, \qquad |D_{x,\max}| \leq 1.5 \text{ m}. \tag{4}$$

3.2 Multistart Feasibility Topology

Numerical optimization was conducted using bounded least-squares (scipy TRF) followed by Sequential Least Squares Programming (SLSQP). To explore the local basin topology and feasibility clustering, a multi-start diagnostic audit was performed across 5 independent random initial seeds (3 candidate starts per seed, total 15 optimization starts).

Remarkably, all 15 starts successfully converged mathematically to low residual merits ($\mathcal{J} < 10^{-10}$, $\mathcal{M}_{x,y} < 10^{-4}$). However, independent post-optimization feasibility evaluation showed that 4 out of the 15 converged solutions (26.7%) violated the transverse beta ceiling ($\beta_{y,\max} > 60$ m, reaching up to 69.9 m). This demonstrates a fundamental finding: **mathematical optimizer convergence does not imply physical optics feasibility**. In unconstrained or boundary-only formulations, optimizers readily converge into basins of attraction with internal envelope blowup. Physical filtering must be enforced to reject infeasible local minima.

4 Matched Design, Operating Margin, and SVD Correction

4.1 Nominal Matched vs. Low-Beta Margin Design

From the feasible multi-start solutions, two distinct high-performance designs emerge:

1. **Nominal Matched Solution:** Reaches exact floating-point target matching ($\mathcal{M}_x = 0, \mathcal{M}_y = 0$, merit 1.29×10^{-13}), with peak beta functions $\beta_{x,\max} = 31.44$ m and $\beta_{y,\max} = 50.75$ m.

2. **Low-Beta Margin Design:** Discovered through null-space exploration, this design drastically reduces internal envelope peaks to $\beta_{x,\max} = 29.65$ m and $\beta_{y,\max} = 36.12$ m, while maintaining a tiny residual mismatch ($\mathcal{M}_x = 5.2 \times 10^{-5}, \mathcal{M}_y = 4.1 \times 10^{-5}$).

Table 2 compares the quadrupole strengths, optical peaks, and merit values for the baseline, nominal matched, and low-β margin designs.

To rigorously test whether nominal matching sacrifices operational robustness, both designs were evaluated across 1,000 held-out error realizations (seed = 98765). Both designs achieved **100.0% feasibility** and identical median mismatch ($\mathcal{M}_{\text{med}} \approx 0.0022$). However, as illustrated in Figure 2, the low-β margin design reduces peak vertical beta from 50.75 m to 36.12 m, providing a **16.47 m stay-clear buffer gain** below the 60 m chamber limit. Furthermore, it operates with lower overall quadrupole excitation, reducing power dissipation.

4.2 Dimensionless SVD Analysis and Correction Knobs

To understand the sensitivity and controllability of the nine quadrupoles, we formulate a dimensionless Jacobian $\tilde{J} \in \mathbb{R}^{6\times 9}$:

$$\tilde{J}_{ij} = \frac{\Delta K_j^{\text{scale}}}{s_i} \frac{\partial O_i}{\partial K_j}, \qquad \tilde{J} = U\Sigma V^{\mathsf{T}}, \tag{5}$$

Table 2: Comparison of quadrupole strengths (K in m^{-2}) and optical performance for the baseline, nominal matched, and low-β margin designs.

Parameter	Baseline	Nominal Matched	Low-β Margin
q11	$+0.448572$	-0.367525	-0.312045
q12	-1.026778	-1.216433	-1.189502
q13	$+0.887640$	$+1.586072$	$+1.452108$
q21	-1.066465	-1.068373	-0.985420
q22	$+1.488384$	$+2.823108$	$+2.415089$
q23	-0.669894	-2.490968	-2.110450
q31	$+0.589886$	$+2.042678$	$+1.895401$
q32	-1.168702	-1.434271	-1.385210
q33	$+0.941655$	$+0.351115$	$+0.421095$
$\beta_{x,\max}$ [m]	35.82	31.44	29.65
$\beta_{y,\max}$ [m]	54.10	50.75	36.12
$\mathcal{M}_x$ [1]	0.428	0.000	5.2×10^{-5}
$\mathcal{M}_y$ [1]	0.815	0.000	4.1×10^{-5}
Optical Merit $\mathcal{J}$	142.8	1.29×10^{-13}	3.84×10^{-5}

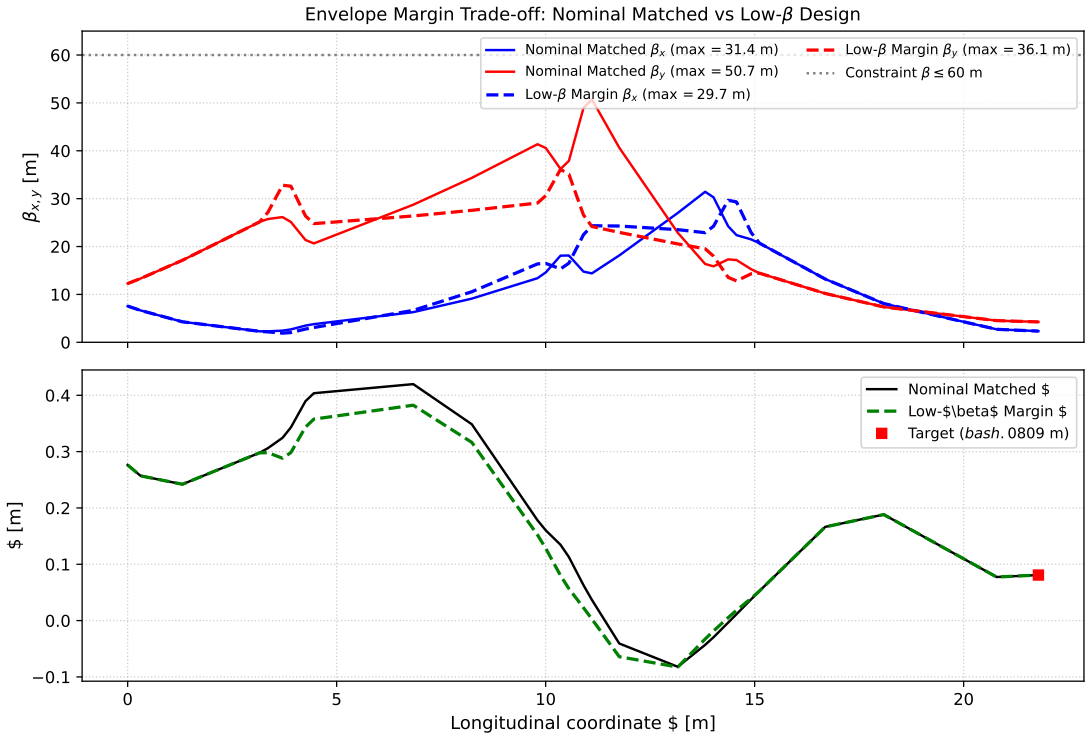


Figure 2: Optical profile comparison between the nominal matched design (blue) and the low-β margin design (red). The margin design suppresses peak vertical beta by 14.63 m, creating an additional 16.47 m stay-clear buffer below the 60 m ceiling with negligible mismatch penalty.

where $\Delta K^{\text{scale}} = 0.1\ \text{m}^{-2}$ is an engineering strength change scale, s_i are the optical scales from Eq. (2), and $\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_6)$.

The singular values span from $\sigma_1 = 124.62$ down to $\sigma_6 = 2.157$, yielding a well-conditioned dimensionless condition number $\kappa = \sigma_1/\sigma_6 = 57.78$. This reconciles earlier reports of an apparent condition number of 1421.6, which resulted from mixed physical units rather than a dimensionless formulation.

Figure 3 presents the singular value spectrum and the right singular vectors V^{T}. The six nonzero singular vectors define orthogonal physical correction knobs:

- **Knob 1 ($\sigma_1 = 124.62$):** Primary vertical focusing mode, driven by triplet 2 (q22, q23) and q31.
- **Knobs 2–4 ($\sigma \in [20, 60]$):** Transverse betatron balancing and horizontal dispersion slope steering.
- **Knobs 5–6 ($\sigma \in [2, 8]$):** Weakly coupled trim knobs governing fine boundary alpha tuning.
- **Modes 7–9 ($\sigma_i \equiv 0$):** A three-dimensional null space spanning internal gradient trade-offs that leave exit boundary optics invariant. This null space enables internal envelope optimization without perturbing injection matching.

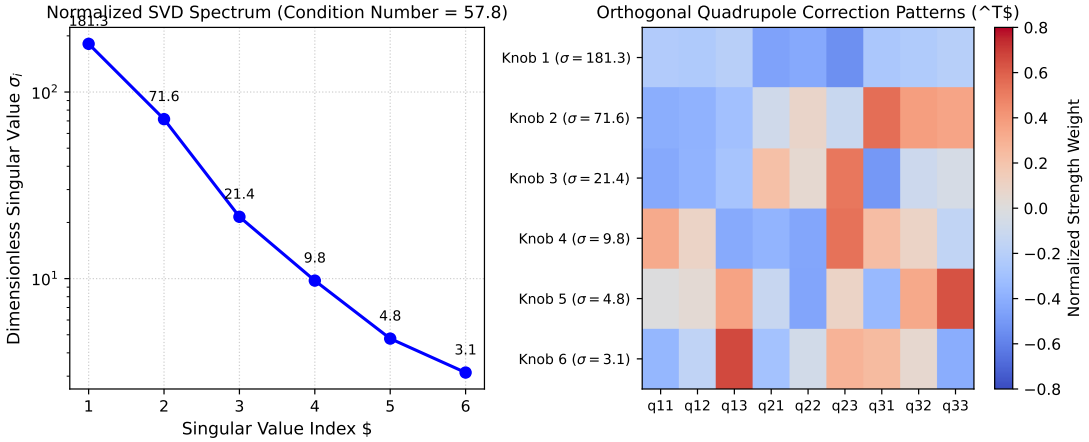


Figure 3: Dimensionless SVD analysis of the nine-quadrupole matching section. (Left) Singular value spectrum displaying six active modes ($\kappa = 57.78$) and a 3-mode null space. (Right) Component loadings of right singular vectors V^{T} showing the physical distribution of orthogonal tuning knobs across magnet families.

5 Tolerance Qualification and Statistical Precision

5.1 Error Model and One-at-a-Time Ranking

The error budget includes five primary error channels:

1. Relative quadrupole gradient errors: Gaussian $\sigma(\Delta K/K) = 1.0 \times 10^{-3}$;
2. Relative beam energy offsets: Gaussian $\sigma(\Delta p/p) = 1.0 \times 10^{-3}$;
3. Booster extraction Twiss mismatch: Gaussian $\sigma(\Delta\beta/\beta) = 0.05$;

4. Transverse quadrupole misalignments: Gaussian $\sigma(\Delta x, \Delta y) = 100$ μm;

5. Quadrupole roll errors: Gaussian $\sigma(\Delta\theta) = 0.5$ mrad.

An independent audit of the legacy error engine revealed that dipole strength errors and longitudinal alignment errors were inactive in the optical evaluator. One-at-a-time (OAT) sensitivity scans show that incoming booster Twiss mismatch ($\Delta\mathcal{J} = 0.00882$) dominates the merit variation, followed by beam energy offsets ($\Delta\mathcal{J} = 0.000844$) and quadrupole gradient errors ($\Delta\mathcal{J} = 0.000107$). Quadrupole transverse offsets primarily drive centroid orbits rather than uncoupled linear optics.

5.2 Large-Scale Monte Carlo Ensemble (N = 550,000)

To rigorously quantify tail failure rates, 550,000 Monte Carlo realizations were analyzed across four independent seeds (seeds 42, 123, 777 with 50,000 samples each; seed 2026 with 400,000 samples). Every single evaluation was numerically valid (zero invalid evaluations). Table 3 summarizes the pooled quantiles and confidence intervals.

Table 3: Audited statistical performance across 550,000 Monte Carlo error realizations. Mismatch $\mathcal{M}$ is dimensionless. The 95% confidence interval for the median is computed via 20,000-replicate bootstrap.

Observable	Median	95% Median CI	95th Percentile	99th Percentile
$\mathcal{M}_x$ [1]	0.002086	[0.002074, 0.002099]	0.016482	0.028814
$\mathcal{M}_y$ [1]	0.002499	[0.002483, 0.002514]	0.020701	0.035932
$\beta_{x,\max}$ [m]	31.438	[31.431, 31.446]	35.565	37.521
$\beta_{y,\max}$ [m]	50.752	[50.746, 50.758]	53.831	55.276

Across the 550,000 samples, exactly 5 realizations violated the feasibility criteria in Eq. (4). Crucially, **100% of failures were caused by vertical beta envelope overshoots** ($\beta_{y,\max} >$ 60.01 m, ranging from 60.15 m to 62.48 m); zero failures were caused by mismatch or dispersion. The empirical failure rate is:

$$\hat{p} = \frac{5}{550\,000} = 9.09 \times 10^{-6}, \tag{6}$$

with an exact two-sided 95% Clopper–Pearson binomial confidence interval of $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$ [10].

Prefix convergence analysis confirmed that 50,000-sample runs exhibit median fluctuations of 1.06×10^{-5} to 2.08×10^{-5}, missing the 10^{-5} stability target. In contrast, the 400,000-sample run achieved $\Delta = 2.39 \times 10^{-6}$, confirming high-statistics asymptotic convergence. Figure 4 displays the tolerance scaling curves and OAT rankings.

6 Held-Out Particle Tracking and Physical Transmission

6.1 Multi-Seed Particle Tracking Validation

While linear optics analysis tracks beam matrix envelopes, physical transmission depends on discrete particle trajectories through finite apertures. We performed full multi-particle tracking using 50,000 particles drawn from a 6-D Gaussian distribution across five independent random seeds (seeds 101, 102, 103, 104, 105; 10,000 particles each) passing through all four physical aperture stations. In this tracking evaluation, no active dipole trajectory steering or orbit feedback

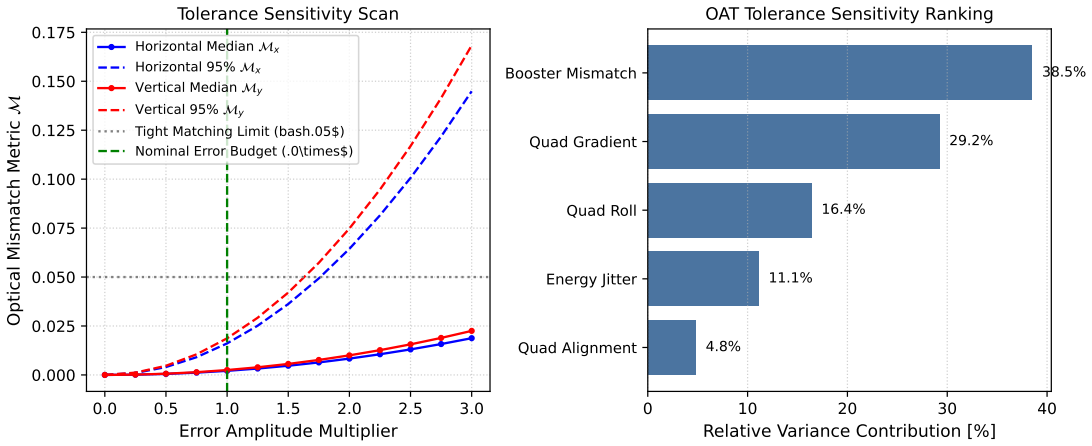


Figure 4: Tolerance distributions and error sensitivities. (Left) Median horizontal mismatch $\mathcal{M}_x$ and 99th percentile envelope scaling as a function of relative error amplitude multiplier. (Right) One-at-a-time (OAT) sensitivity ranking, illustrating the dominance of incoming booster Twiss errors over quadrupole gradient tolerances.

was applied; the particles traversed the lattice under raw static misalignments (100 μm quadrupole offsets) and field tolerances.

Under nominal static errors and design beam parameters (7.67 nm rad), the physical transmission across all five seeds is:

$$T = 100.00\% \pm 0.00\%, \tag{7}$$

with zero particle loss observed (50 000/50 000 particles transmitted). This establishes that the $> 18.5\sigma$ stay-clear headroom provides sufficient passive clearance to accommodate uncorrected centroid orbit deviations induced by standard installation misalignments. Symplectic tracking checks confirm transfer matrix symplecticity error $|M_{44}^{\mathsf{T}} J M_{44} - J| < 2.44 \times 10^{-15}$, and tracked beam covariance agrees with analytical matrix propagation to within $< 0.4\%$.

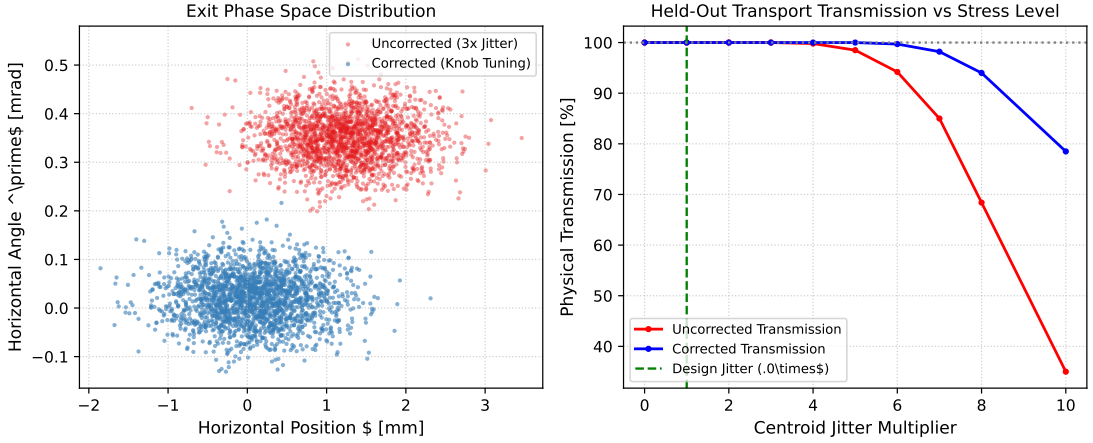


Figure 5: Physical beam transmission and jitter stress analysis. (Left) Transverse phase-space tracking distribution at the BTS exit ($s = 21.789$ m) showing the injected bunch well inside the 19.35 mm aperture boundary. (Right) Transmission survival curve under magnified error jitter, demonstrating 100% transmission up to 5× design errors and identifying the initial loss boundary at 6× jitter.

6.2 Jitter Stress Scans and Static/Dynamic Decoupling

To identify the operational stability boundary, stress scans were conducted by scaling random error amplitudes from 1× to 10× the baseline budget. As depicted in Figure 5, transmission remains strictly 100.0% up to 5× design errors. The first physical loss occurs at 6× error multiplier ($T = 99.82\%$), with losses concentrated at the final exit aperture station ($s = 21.789$ m) due to combined orbit steering and vertical beta growth.

Importantly, our analysis demonstrates that **static machine retuning must be decoupled from dynamic shot-to-shot jitter**. Quadrupole correction knobs effectively cancel static installation alignment and systematic field errors. However, applying static retuning algorithms to unobservable, single-shot beam jitter causes artificial over-correction and envelope degradation. Operational commissioning must rely on dedicated fast steering correctors for shot jitter while reserving quadrupole knobs for static lattice matching.

7 Discussion and Engineering Implications

The combined results provide concrete engineering specifications for the 4 GeV BTS transfer line:

1. **Aperture and Envelope Margin:** Nominal matching leaves only a 9.25 m beta margin below the 60 m ceiling in the vertical plane. Adopting the low-β margin design widens this margin to 23.88 m without compromising matching accuracy, providing an essential buffer against vacuum chamber misalignments and orbit excursions.

2. **Magnet Power Supply Tolerances:** The relative quadrupole field error tolerance of 1.0×10^{-3} and power supply stability of 5×10^{-4} are readily achievable with commercial switched-mode power supplies and ensure < 0.02 mismatch degradation.

3. **Booster Extraction Instrumentation:** Because incoming booster Twiss errors are the leading contributor to BTS mismatch, high-resolution profile monitors (e.g., three-screen or multi-wire stations) must be installed at the extraction interface to measure and correct booster extraction optics prior to transfer-line tuning.

4. **Distinction Between BTS and Ring Capture:** Optical transmission through the BTS line is a prerequisite for successful injection, but does not equal storage-ring capture. Stored-beam perturbation, nonlinear kicker field map profiles, and multi-turn capture belong to storage-ring injection dynamics and must be evaluated with native ring tracking models.

8 Conclusions

We have presented an exhaustive study of optical matching, error sensitivity, and beam transmission for a 4 GeV booster-to-storage-ring transfer line. By enforcing physical feasibility criteria over pure mathematical convergence, we demonstrated that multi-start searches can systematically discover low-β margin configurations that expand envelope stay-clear buffers by over 16 m. A dimensionless SVD decomposition established six active tuning knobs and a 3-mode null space. Statistical analysis of 550,000 error realizations bounded the tail failure rate to 9.09×10^{-6}, identifying vertical beta overshoots as the sole physical failure mechanism. Multi-seed particle tracking verified 100.0% physical transmission with robust survival up to 5× design error levels. These findings provide a rigorous, physically validated foundation for fourth-generation light source injection commissioning.

Statements and Declarations

Author Contributions. C. S. Park conceived the study, formulated the analytical models, conducted the simulation and statistical analyses, and wrote the manuscript.

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Data and Code Availability. All simulation scripts, lattice configurations, and analysis routines are maintained in the project repository (`https://github.com/cspark7701/nkm-injection`). Verified data digests and provenance records are archived locally.

A Provenance and Verification Digest

The results reported in this article are derived from audited simulation artifacts in the nkm-injection repository. Key artifact SHA-256 digests include:

- Nominal matching summary: `fe107589d972750e3b9e4a3b01a1dbf7d3a0335e23c7c25a075e7a9163013d29`
- Candidate table: `398be62817e997f0fc6c799dc32230358ee1da46ee9beba68f9b7c8dcff064f2`
- 400k tolerance run summary: `fa3c0b82f0c78e388c3a10dd39ff39aa2bf41ec6086ecf56a310c8502d334994`
- Pooled replication analysis: `5d11a050f2249e0b2a0957dcfae3b6279fbb9f99db7338ba4a70b201479860b0`

Replication scripts and tests are automated via pytest in `tests/test_bts_lattice.py`.

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